UNIVERSITY OF UYO

DEPARTMENT OF MATHEMATICS

FIRST SEMESTER EXAMINATION SESSION: 2024/2025

COURSE CODE: MTH411(ORDINARY DIFFERENTIAL EQUATION I) TIME: 3HRS

INSTRUCTION: ANSWER ANY FOUR (4) QUESTIONS ONLY

1a Let $f: D \subset \Re^n \to \Re^n, n \geq 1$, and $y:[a,b] \to \Re^n$ be continuous on the closed set $D = \{x, y : \|x - x_0\| \leq a, \|y - y_0\| \leq b, a, b \in \Re\}$ so that

$$y' = f(x, y), \quad y(x_0) = y_0 . \qquad (1.1)$$

i. Show that $y = y_0 + \int_{x_0}^{x} f(s, y(s))ds$ satisfies equation (1.1)

ii. Discuss the existence and unique solution of the initial-value problem

$$y' = \frac{2y}{x}, \qquad y(x_0) = y_0$$

b. Define the smallest interval that guarantees the existence and uniqueness of the solution of the initial-value problem

$$y' = 1 + y^2, \quad y(0) = 0,$$

2a. Show that if $X(t)$ is the fundamental matrix of homogeneous part of system

$$y' = Ay(x) + f(x), \quad y(x_0) = y_0, \qquad (2.1)$$

then the general solution of the initial value problem (2.1) is

$$y(x) = X(x)X^{-1}(x_0)y_0 + \int_{x_0}^{x} X(x)X^{-1}(s)f(s)ds$$

b. Find the fundamental matrix solution of the differential equation $y' = \begin{pmatrix} 4 & -3 \\ 8 & -6 \end{pmatrix} y$

3. Find the particular solution of the initial value problem

$$y' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} y + \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad y(0) = (1, \ 1)^T$$

4ai. What is the 2nd order scalar equation equivalent to the first order system $\begin{aligned} y_1' &= 3y_1 - 2y_2 \\ y_2' &= 2y_1 - 2y_2 \end{aligned}$.

ii. Show that the transformed equation is actually equivalent to the system.

b Determine if the differential equation $x^2 \frac{d^2y}{dx^2} + 7x\frac{dy}{dx} + 8y = 0$ is a self-adjoint equation. If it is not, transform the equation into a self-adjoint equation.

5a. Given $\varphi(t)$ is the fundamental matrix solution of the homogeneous linear system with periodic coefficient

$$y' = A(t)y, \quad A(t + w) = A(t), \qquad (5.1)$$

for some $w > 0$, $t \in I \subset \Re$.

What conditions guarantee the existence of the periodic solution of equation (5.1)?.

bi. Show that the equation $y' + p(x)y = r(x)$ satisfies Lipschitz theory on uniqueness

ii. Define the type of convergence for each of the following functions